COMPSCI5079(M)
Cryptography & Secure Development

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Lecture 2:

Traditional Encryption and Decryption schemes

Lecture Outline

1. Types of Attack
   - Threats that need to be protected against.
2. Transposition and Substitution ciphers.
3. Homophonic Ciphers
   - Plaintext letters can have several ciphertext letters.
4. Polyalphabet Ciphers and Rotor Machines.
   - Mechanical encryption devices.
5. Running key Ciphers
   - The key is the same length as the plaintext

Homophonic Ciphers

Homophonic Cipher

- A homophonic cipher matches each letter in the plaintext alphabet to possibly more than one letter in another alphabet used for the ciphertext.

- For example, English letters in the plaintext to numbers between 0 and 127 in the ciphertext.
  - The ciphertext alphabet is numbers between 0 and 127.

- Each English letter in the plaintext can correspond to several numbers in the ciphertext.
  - The key is this match between letters and numbers.
  - The key can be quite large.

Homophonic Cipher

- More frequently used letters have more matching characters in the other alphabet.
  - This severely weakens attacks based on single-letter frequencies
- It can still be attacked by "digram frequencies".

- The following slide shows how many numbers between 0 and 127 should be allocated to each of the letters in the alphabet, based on the earlier `dracula.txt` frequencies.

- It also shows the relative frequency of plaintext letters to ciphertext codes.
  - 1.00 means no statistical information can be obtained.

- Note that just the rare letters contain statistical information, and they are rare, which does not help much.

Codes per Letter

a	---	10	1.05	n	---	8	1.09
b	---	2	0.90	o	---	10	1.01
c	---	3	0.90	p	---	2	0.92
d	---	6	0.95	q	---	1	0.13
e	---	16	0.99	r	---	7	1.00
f	---	3	0.94	s	---	8	0.99
g	---	2	1.27	t	---	11	1.06
h	---	8	1.08	u	---	4	0.90
i	---	8	1.07	v	---	1	1.18
j	---	1	0.16	w	---	4	0.91
k	---	1	1.24	x	---	1	0.16
l	---	5	1.05	y	---	2	1.27
m	---	3	1.19	z	---	1	0.07

Homophonic Cipher

- The larger the ciphertext alphabet, the more secure the code.
- In the limit where every plaintext letter is encrypted to a different ciphertext letter, the code cannot be broken
  - However, the key size is at least as large as the length of the plaintext.
- The ciphertext alphabet requires more bits to encode each letter.
  - 26 letters require 5 bits per letter
  - Numbers between 0 and 127 require 7 bits per number.

Homophonic Cipher: Example

Example: English letters are enciphered as integers (0 - 99), a group of integers are assigned to a letter proportional to the relative frequency of the letter, as in the table:

M = PLAIN PILOT (plaintext)
C = 91 44 56 65 59 33 08 76 28 78 (ciphertext)

Letter	Homophones (Codes)
A	17 19 34 4 56 60 67 83
I	08 22 53 65 88 90
L	03 44 76
N	02 09 15 27 32 40 59
O	01 11 23 28 42 54 70 80
P	33 91
T	05 10 20 29 45 58 64 78 99

Second-Order Homophonic

- It is possible to encrypt two different plaintext messages of the same length with two different keys to produce a composite ciphertext.

- Each key will decrypt the ciphertext to produce a different message.
  - The first key produces the real message.

- The other message can be innocuous, and the second key is a ***distress key***, to be revealed under duress.
  - Often called the 'rubber hose' decryption technique.

Second-Order Homophonic: Example

- The plaintext alphabet forms the rows and columns of an $n \times n$ matrix.
  - n is the number of letters in the plaintext alphabet.
- The ciphertext alphabet consists of integers between 0 and n^2-1 inclusive
  - There are n^2 of them.
- The key is the order in which the ciphertext letters appear in the matrix.
- Encryption uses letters from the first message to locate the row (Row F1) and the second message for the column (Column F2).
  - Thus, each ciphertext character is the number appearing in the appropriate element in the matrix.

Second-Order Homophonic

For the Decryption:

- Decrypting the first message decodes all the numbers in a **given row** to the same letter

- Decrypting the second message uses the numbers in a **given column** to produce the same letter.

Second-Order Homophonic: Example

Alphabet of 5 letters EILMS

- Encryption Matrix:
- The numbers in the matrix are randomly selected.

	E	I	L	M	S
E	10	22	18	02	11
I	12	01	00	05	20
L	19	06	23	13	07
M	03	16	08	24	15
S	17	09	21	14	04

- If the real message is SMILE and the decoy (dummy) message LIMES, then the ciphertext will be:

  21, 16, 05, 19, 11

Second-Order Homophonic: Example

- The real `decryption key` will be:

```
E(10,22,18,02,11)
I(12,01,00,05,20)
L(19,06,23,13,07)
M(03,16,08,24,15)
S(17,09,21,14,04)
```

- while the `decoy key` will be:

```
E(10,12,19,03,17)
I(22,01,06,16,09)
L(18,00,23,08,21)
M(02,05,13,24,14)
S(11,20,07,15,04)
```

- Note that letter frequencies are not destroyed.

Quizzes

- An isolated civilisation has developed a written language based on an alphabet with just 4 letters: α, β, γ and δ. Their written documents are very long.
  - The letters do not occur with equal frequency: α occurs 3/8 of the time ; β and γ 1/4 of the time each and δ 1/8 of the time.
  - The probability of any two-letter combination occurring is, however, just the product of the probability of each letter occurring independently and there are no special digrams or trigrams.

Quiz: Calculate the redundancy of this language.

You do not need to calculate an exact number but can leave terms like $\log_2(3)$ in your answer.

Quizzes

- This civilisation is aware of the English language and has decided to encrypt its secret documents by using some English language letters for the ciphertext.

Quiz: Show how they can hide the redundancy in their language by using the alphabet {A, B, C, D, E, F, G, H} as the ciphertext alphabet.

- This civilisation is also investigating the possibility of hiding two different messages in the ciphertext, each with its own key.

Quiz: How many letters from the English language would be needed? Give an example of such an encoding. Explain why this code would be easier to break than the previous one.

Lecture Summary

Homophonic Cipher

- A homophonic cipher has more letters in the ciphertext alphabet than in the plaintext alphabet.
  - Each plaintext letter can be replaced by a choice of several ciphertext letters (chosen at random).
  - This reduces the effect of statistics.

- A second-order homophonic encrypts two different messages, each with its own key, into one piece of ciphertext.
  - An innocent message is protected by a distress key.